



SKIN TYPE DETECTION AND SKINCARE RECOMMENDATION

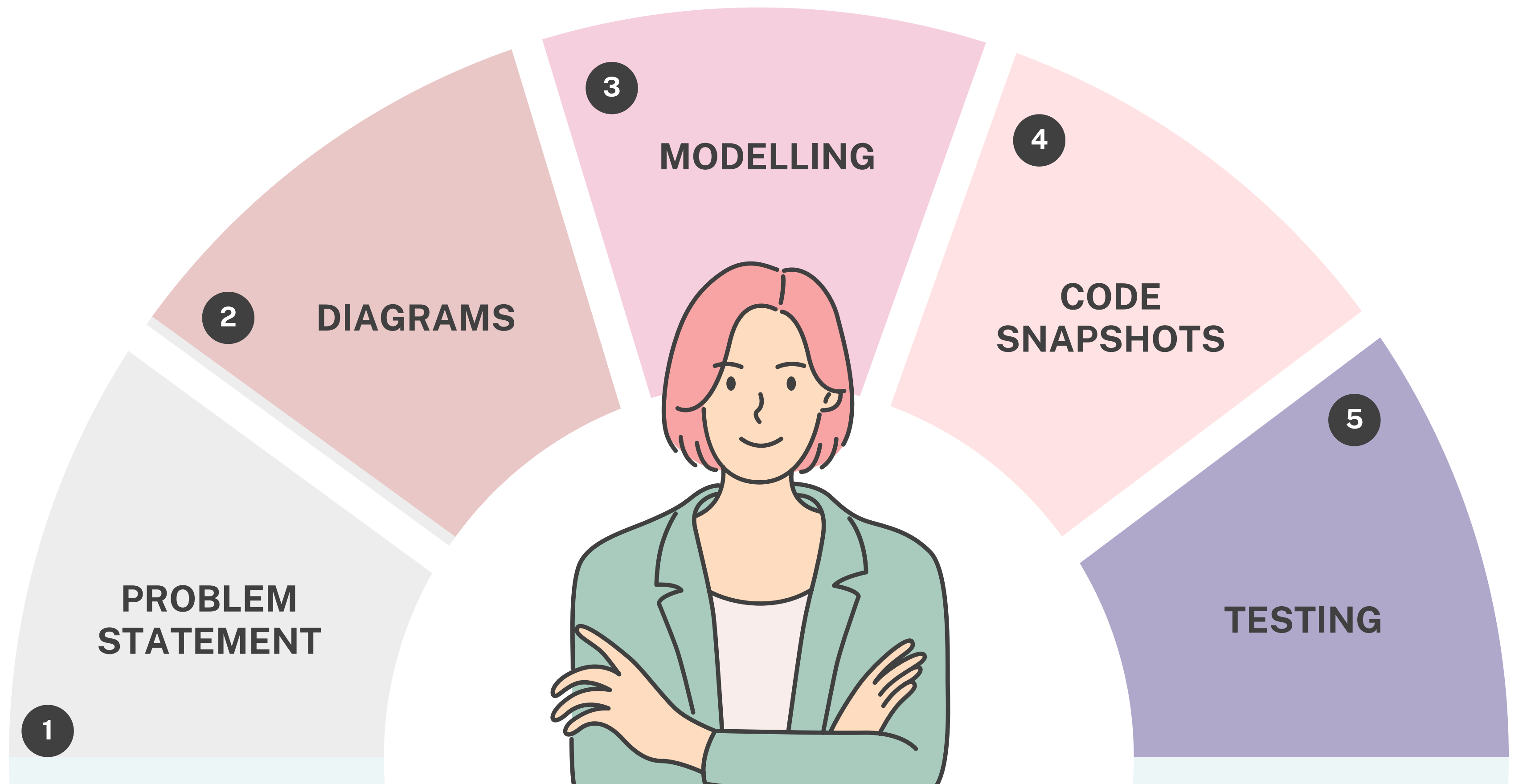
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INTRODUCTION

- This project focuses on developing an intelligent system that identifies an individual's skin type using advanced deep learning techniques and subsequently recommends appropriate skincare products by leveraging web scraping technologies.
- Upon determining the skin type, our system proceeds to recommend tailored skincare products. This is achieved through web scraping, a technique that extracts relevant product information from various online skincare retailers.
- By compiling data on product ingredients, user reviews, and expert recommendations, our model ensures that the suggestions are both scientifically sound and user-approved.



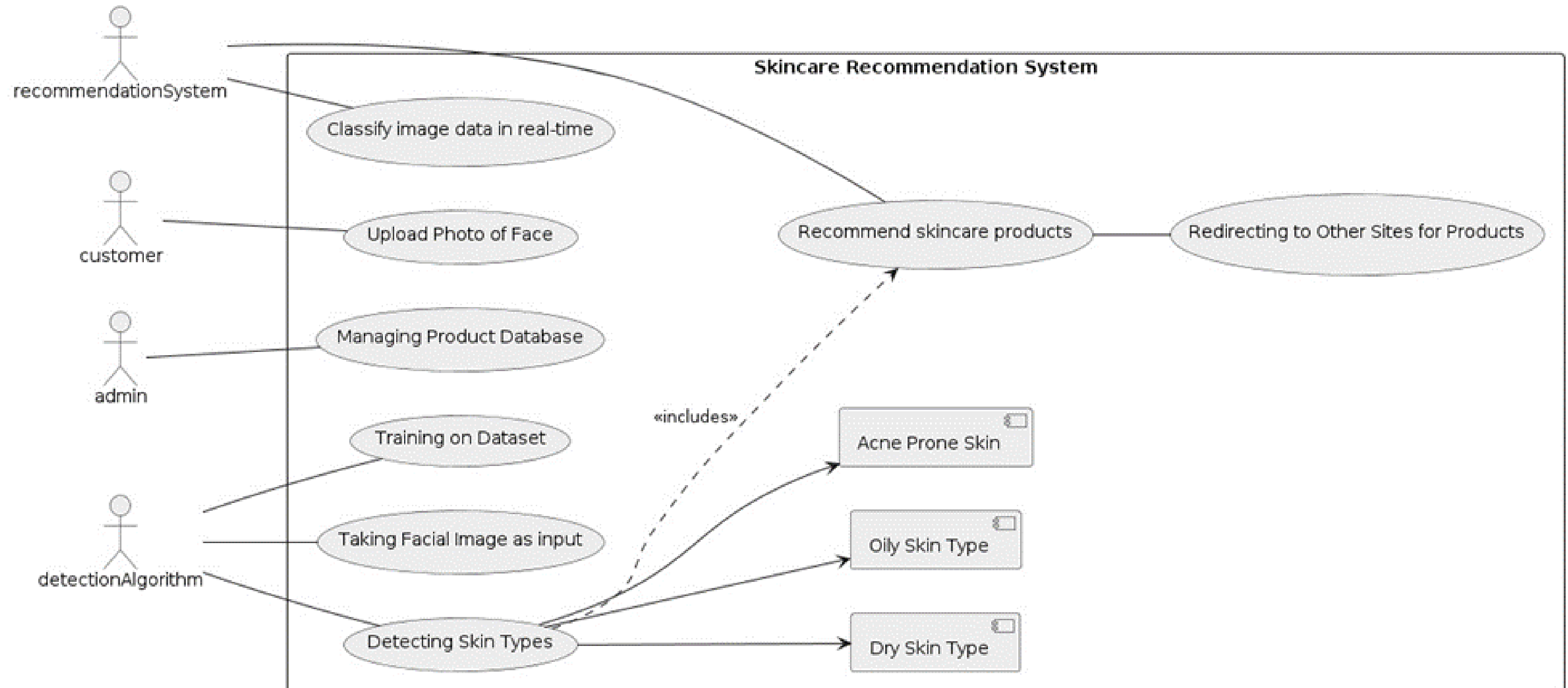
PROBLEM STATEMENT

People often struggle to find the right skincare products for their specific skin types due to the overwhelming variety of choices and the lack of personalized advice.

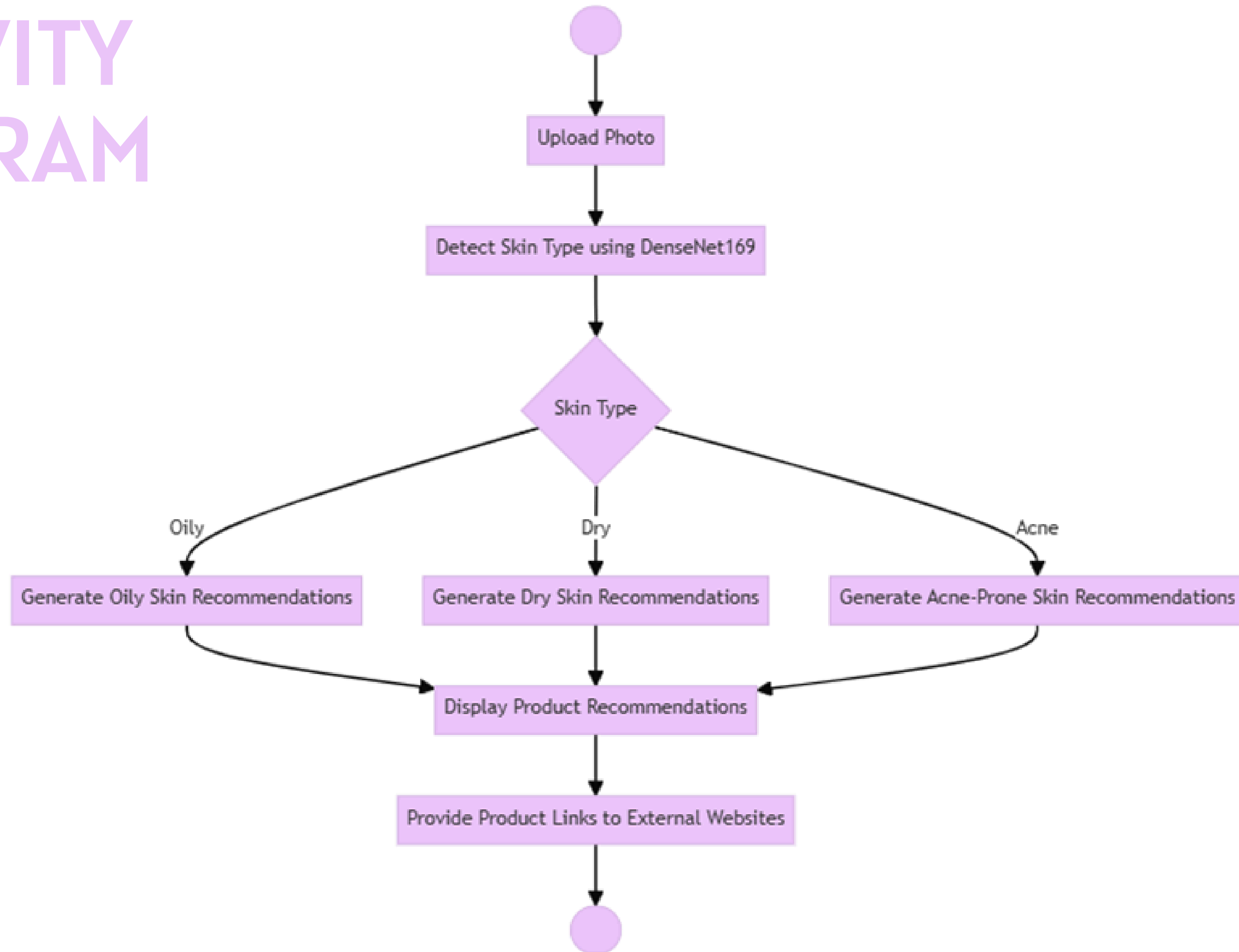
Current solutions either require professional dermatological consultations, which can be costly and time-consuming, or rely on generic recommendations that do not account for individual variations. This leads to ineffective skincare routines, wasted resources, and frustration.

DIAGRAMS

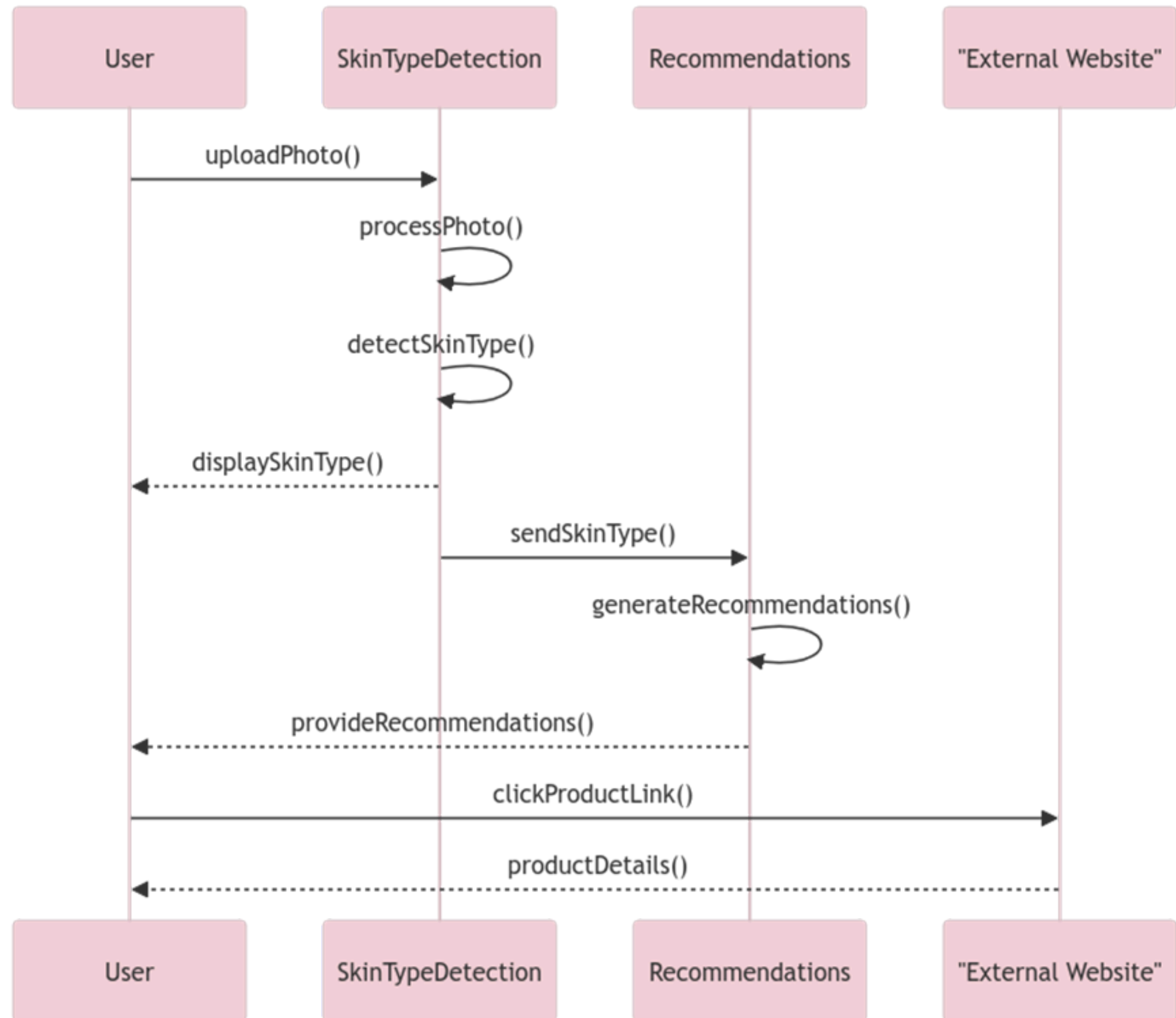
USE CASE DIAGRAM



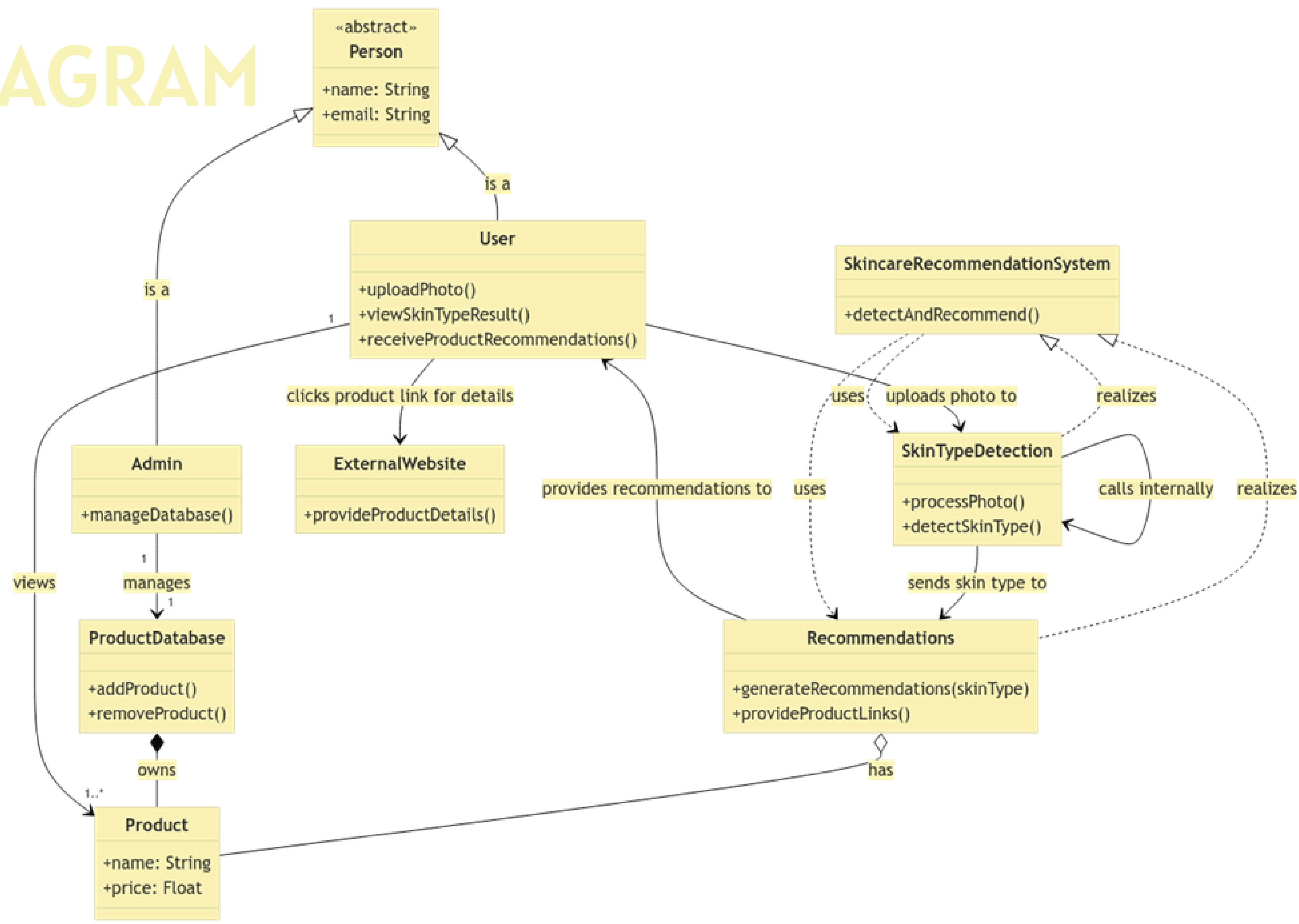
ACTIVITY DIAGRAM



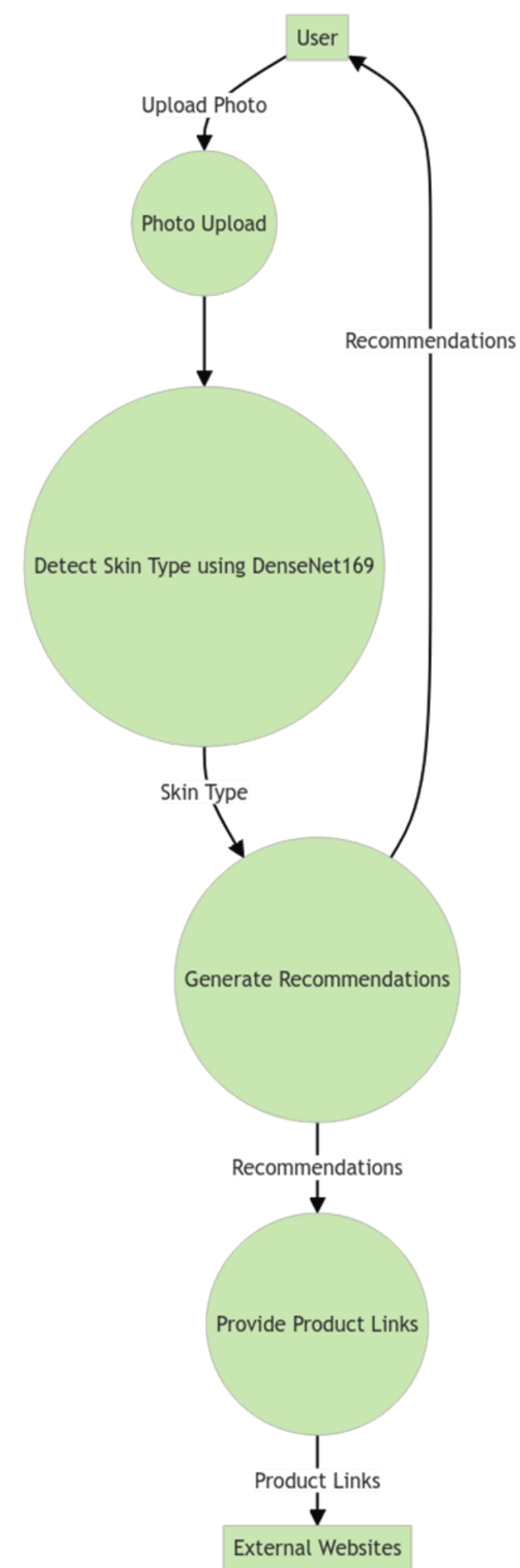
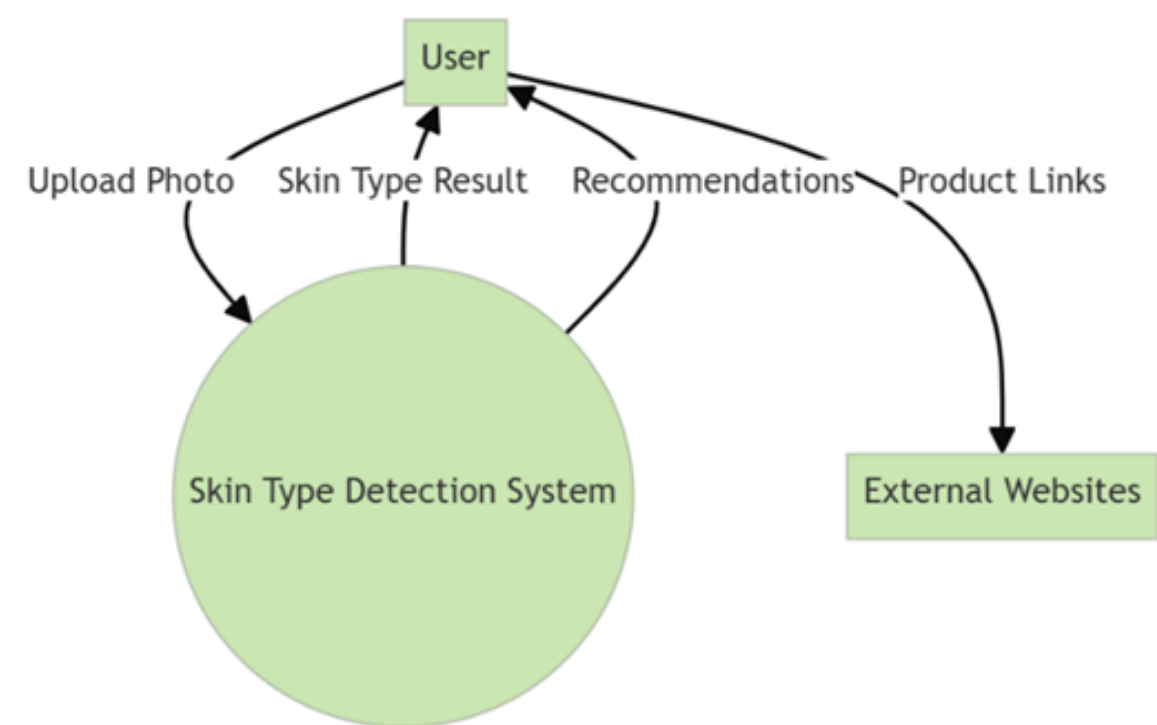
SEQUENCE DIAGRAM



CLASS DIAGRAM



DATA FLOW DIAGRAM



CODE SNAPSHOTS

MODELLING

```
from tensorflow.keras.layers import Dropout, BatchNormalization
from tensorflow.keras.models import Sequential
keras_model = Sequential()
# Add the pre-trained DenseNet169 model
pretrained_model = tf.keras.applications.DenseNet169(include_top=False,
                                                    input_shape=(640, 640, 3),
                                                    pooling='avg', weights='imagenet')

# Freeze the layers of the pre-trained model
for layer in pretrained_model.layers:
    layer.trainable = False
keras_model.add(pretrained_model)
keras_model.add(Flatten())

# Add a dense layer with dropout and batch normalization
keras_model.add(Dense(512, activation='relu'))
keras_model.add(BatchNormalization())
keras_model.add(Dropout(0.5))

# Add another dense layer
keras_model.add(Dense(256, activation='relu'))
keras_model.add(BatchNormalization())
keras_model.add(Dropout(0.5))

# Add the final classification layer
keras_model.add(Dense(3, activation='softmax'))

# Compile the model
keras_model.compile(optimizer=Adam(learning_rate=0.001),
```

ARCHITECTURE

Layer (type)	Output Shape	Param #
densenet169 (Functional)	(None, 1664)	12642880
module_wrapper (ModuleWrapper)	(None, 1664)	0
module_wrapper_1 (ModuleWrapper)	(None, 512)	852480
batch_normalization (Batch Normalization)	(None, 512)	2048
dropout (Dropout)	(None, 512)	0
module_wrapper_2 (ModuleWrapper)	(None, 256)	131328
batch_normalization_1 (Batch Normalization)	(None, 256)	1024
dropout_1 (Dropout)	(None, 256)	0
module_wrapper_3 (ModuleWrapper)	(None, 3)	771
Total params: 13630531 (52.00 MB)		
Trainable params: 986115 (3.76 MB)		
Non-trainable params: 12644416 (48.23 MB)		

SKIN TYPE DETECTION

```
# Load and preprocess the image
img_array = load_and_preprocess_image(img_path)
# Make prediction
predictions = model.predict(img_array)
# Get the class labels
labels = list(class_indices.keys())
# Decode the prediction
top_index = np.argmax(predictions[0])
top_label = labels[top_index]
top_confidence = predictions[0][top_index]
return top_label, top_confidence

# Assuming class indices from the test generator
class_indices = {'acne': 0, 'dry': 1, 'oily': 2}

# Path to the image you want to predict
test_image_path = '/content/drive/MyDrive/dataset/acne1.jpeg' # Replace with your test image path

# Predict the class of the test image
predicted_label, confidence = predict_image_class(keras_model, test_image_path, class_indices)

# Print the prediction
print(f"The image belongs to: {predicted_label} with confidence: {confidence:.4f}")

# Optionally, display the image
img = image.load_img(test_image_path, target_size=(640, 640))
plt.imshow(img)
```

SKIN TYPE DETECTION

[18]

```
... 1/1 [=====] - 7s 7s/step  
The image belongs to: acne with confidence: 0.9998
```

...

Predicted: acne (0.9998)



SKINCARE RECOMMENDATION

```
def display_recommended_products(recommended_products,width='100%'):
    #display(HTML(recommended_products.to_html(escape=False)))
    html=recommended_products[['Product_Name', 'Rating', 'Product_Link', 'Description','Price']].to_html(escape=False, index=False)
    styled_html = f'<div style="width: {width}; margin: 0 auto;">{html}</div>'
    display(HTML(styled_html))

def recommend_products(predicted_label, num_recommendations=5):
    if predicted_label == 'acne':
        recommended_products = acne_products.head(num_recommendations)
    elif predicted_label == 'oily':
        recommended_products = oily_products.head(num_recommendations)
    elif predicted_label == 'dry':
        recommended_products = dry_products.head(num_recommendations)
    else:
        raise ValueError("Unknown skin type predicted.")
    return recommended_products

recommended_products = recommend_products(predicted_label)
recommended_products['Product_Link'] = recommended_products['Product_Link'].apply(make_clickable)
display_recommended_products(recommended_products,width='100%')
```

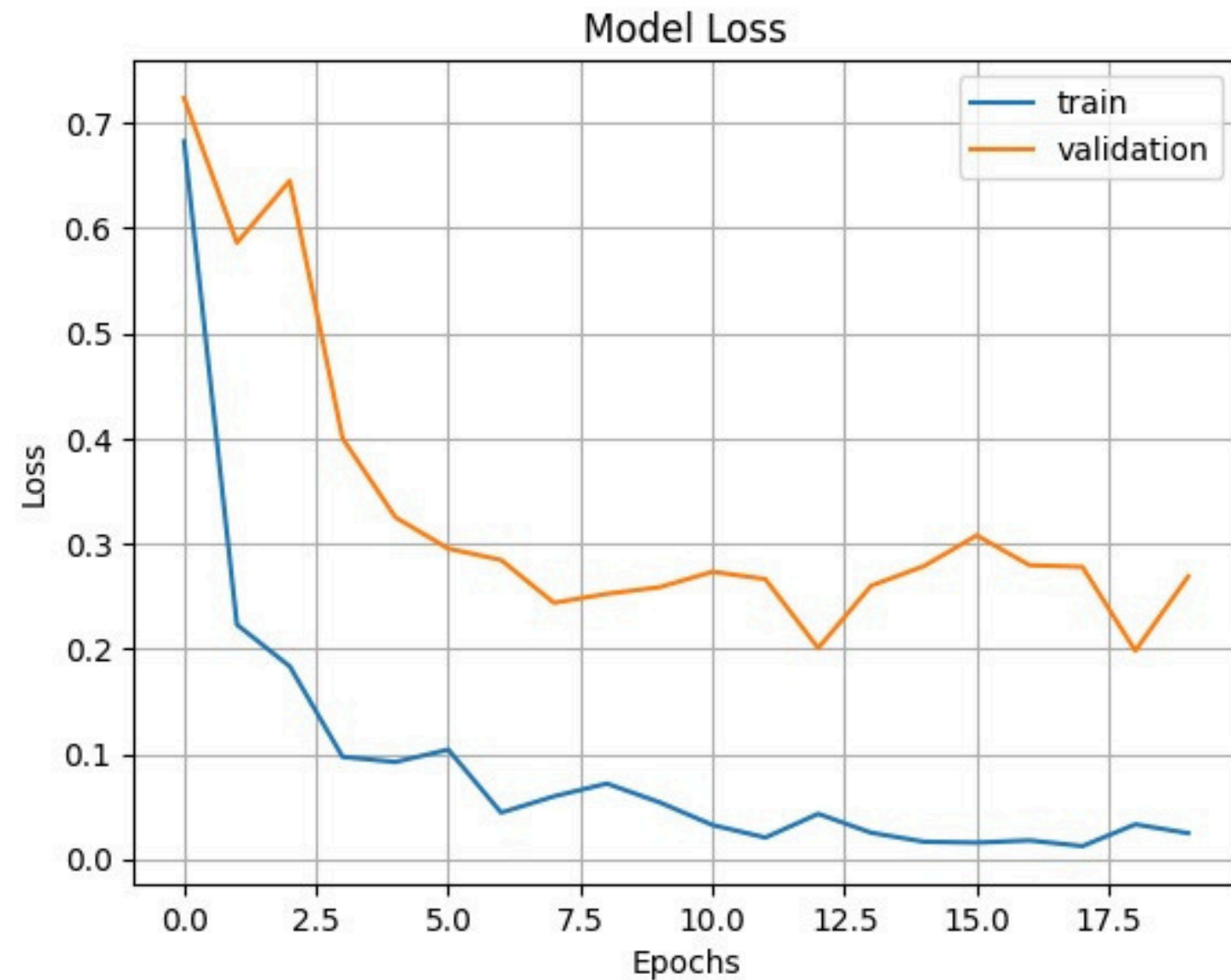
SKINCARE RECOMMENDATION

Product_Name	Rating	Product_Link	Description	Price
Love Beauty & Planet Skin Moisturising Combo At Nykaa, Best Beauty Products Online	5.050	https://www.nykaa.com/love-beauty-planet-skin-moisturising-combo/p/11771942?ptype=product&skuld=11771942&srsltid=AfmBOopv_vsfgzcvmitp_dsSAVUh2ju5BjBs8SrQrPtq_7bhHKruJ7Ks5QM	NaN	533
Physiogel Hypoallergenic Daily Moisture Therapy Cream	5.050	https://www.amazon.in/Physiogel-Hypoallergenic-Moisture-Sensitive-Provides/dp/B0BD29N17Q?source=ps-sl-shoppingads-lpcontext&ref_=fplfs&psc=1&smid=A1ZK88J0676FJ2	Men's · For Dry Skin · Cream	636.5
Dry Skin Care Routine Kit: Face Wash + Face Serum + Face Moisturiser Earth ...	4.927	https://earthrhythm.com/products/dry-skin-care-kit?variant=39474064228396&cy=INR&utm_medium=product_sync&utm_source=google&utm_content=sag_organic&utm_campaign=sag_organic&srsltid=AfmBOoqkrxGEcJ81skayjiRTTTvfspCQDVnTI4yyangdjJR_0YpnJIG1M5k	NaN	₹ 1,587.00
The Derma Co. x Dr.V Skin Renew Barrier Repair Peptide Moisturizer- Peptide ...	4.890	https://www.amazon.in/Derma-Moisturizer-Peptide-Moisturizes-Regeneration-Strengthens/dp/B0D1C19JRP?source=ps-sl-shoppingads-lpcontext&ref_=fplfs&psc=1&smid=A61C6NUFGVGS7	Face · Men's · Vitamin C	499
Re'equil Dry Skin CTM Bundle	4.766	https://www.reequil.com/products/dry-skin-ctm-bundle?variant=32696053137540&cy=INR&utm_medium=product_sync&utm_source=google&utm_content=sag_organic&utm_campaign=sag_organic&srsltid=AfmBOorKZ3HCtOV0fGrnoH13bfDW0Bo4bnP7XVKU2rV-IHrB_6dpfpyr0GE	NaN	1055.55

TESTING

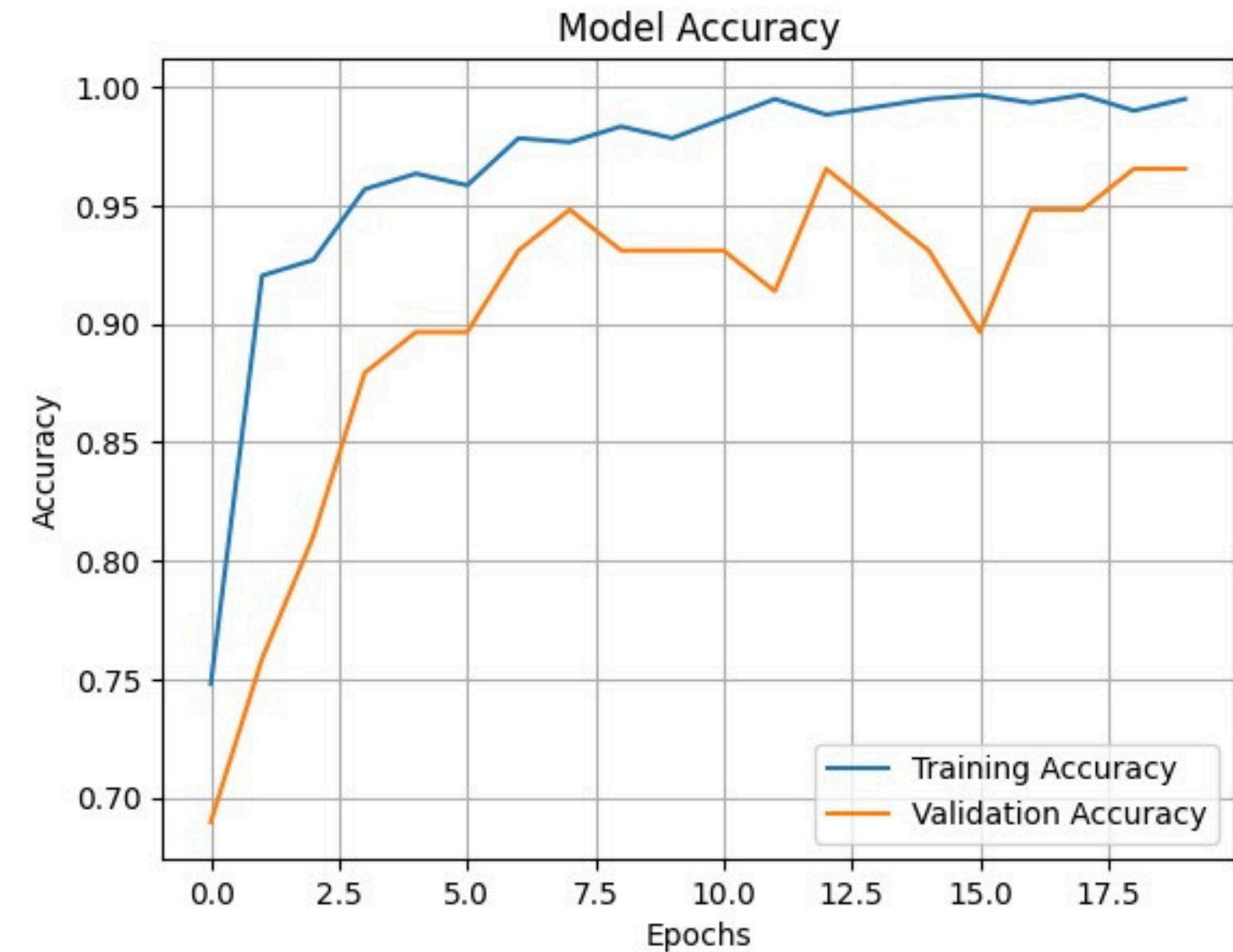
- **Unit Testing:** Verifies code components; ensures specific functionalities.
- **Integration Testing:** Checks system interactions; ensures components work together.
- **Smoke Testing:** Validates basic functionality; confirms software stability.
- **Regression Testing:** Re-runs tests; validates changes without introducing issues.
- **Black-box Testing:** Evaluates functionality without internal knowledge; ensures external behavior meets requirements.
- **White-box Testing:** Tests internal structures with code knowledge; validates internal operations.
- **Performance Testing:** Assesses software speed, responsiveness, and stability under varied conditions.

TESTING



Model Loss

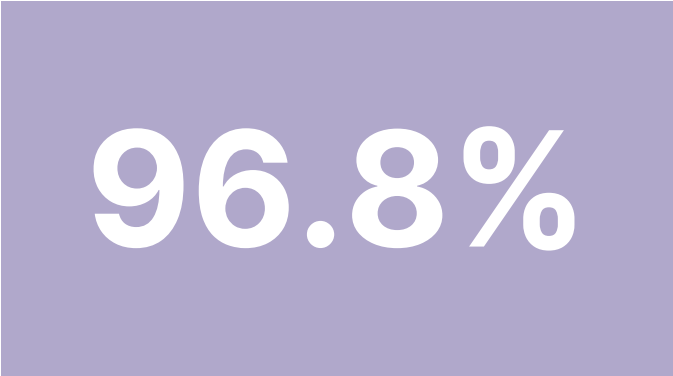
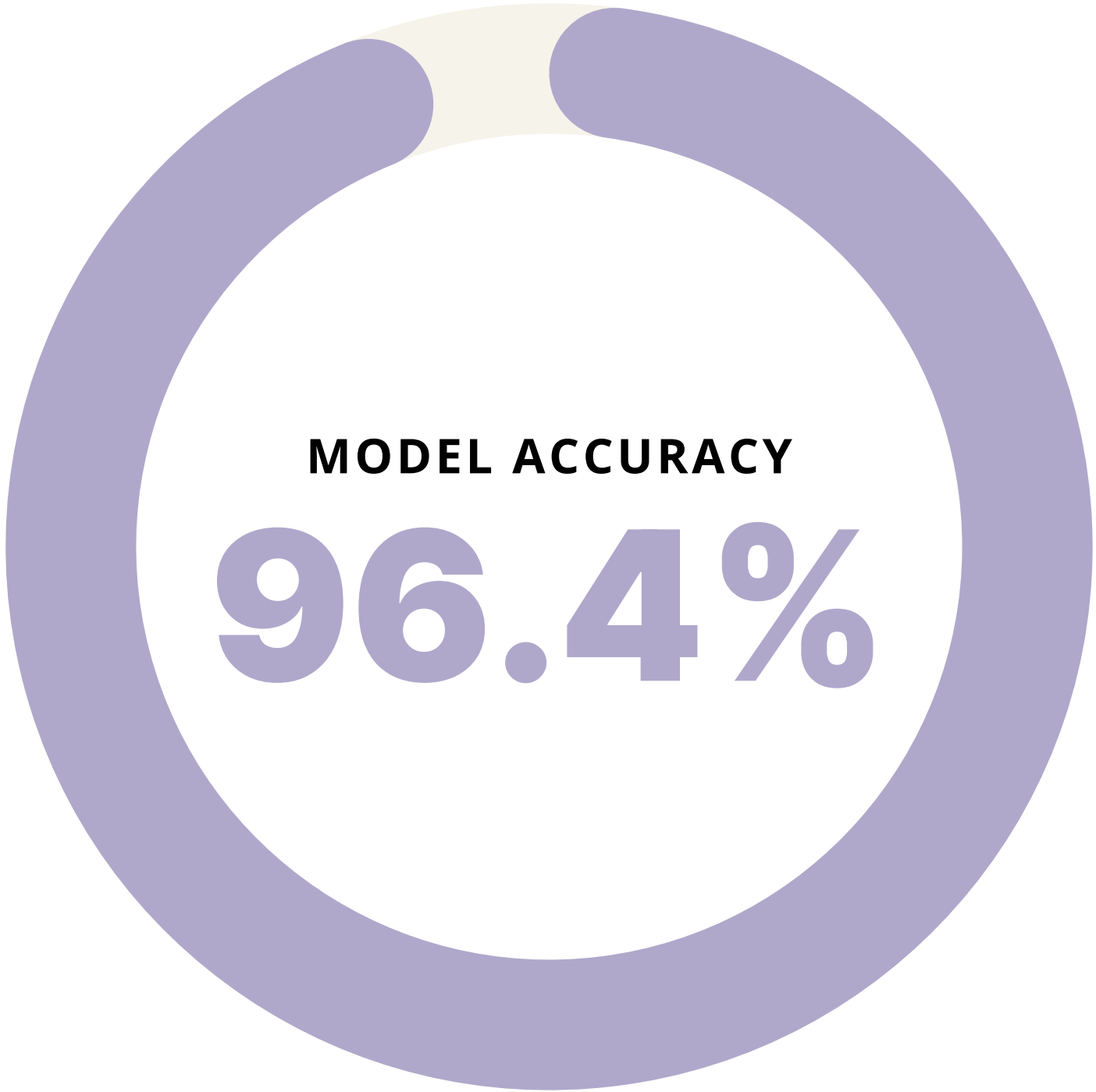
The loss graph represents the performance of the model in predicting skin types or recommending skincare products. It shows the evolution of the loss function, which measures the discrepancy between the predicted and actual skin types or skincare recommendations.



Model Accuracy

The accuracy graph shows how accurately the model predicts skin types or recommends skincare products over the training epochs. It displays the proportion of correct predictions to the total predictions made.

CUSTOMER SATISFACTION





THANK YOU